OAK INTERNATIONAL SCHOOL

EXCELLENCE ENSEMBLE • DUAL CURRICULUM MATHEMATICS

IB MIDDLE YEARS & DIPLOMA	BENIN NATIONALE APC	BILINGUAL EN / FR

PREMIUM MATHEMATICS ACTIVITY WORKBOOK

OAK HARMONISATION STRATEGY & MODEL

A Harmonised Dual-Curriculum Mathematics Learning Companion

Connecting Conceptual Inquiry, Axiomatic Proof, and Examination Mastery

Target Cohort:	Oak Harmonisation Strategy & Model
Academic Year:	2026–2027
Curricular Alignment:	International Baccalaureate & République du Bénin (MEMP/MESTFP)
Publication Code:	OIS-REP-04-STRATEGY
Series Version:	OIS-MTH-STR-2026-V1.0 (Classroom & Exam Certified)

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Deliverable 4: Pedagogical Harmonisation Strategy

Document Reference: OIS-MTH-STR-2026-V1.0

Curricular Scope: Dual-Curriculum Integration: IB (MYP/DP) & Beninese National APC

Strategic Principle: Absolute Curricular Integrity (Zero Compromise of Official Syllabi)

EXECUTIVE SUMMARY & CORE DOCTRINE

At Oak International School, students pursue two rigorous, prestigious mathematical pathways concurrently: the English-medium International Baccalaureate continuum (MYP 1–5 and DP 1–2) and the French-medium Beninese national curriculum under the Approche Par Compétences (APC).

To succeed, the school must avoid two fatal pedagogical pitfalls:

1. **Curricular Dilution**: Omitting required Beninese competencies to fit the IB schedule, or simplifying IB inquiry objectives to match national drill expectations.
2. **Cognitive Fragmentation**: Teaching the two programmes as disconnected silos, subjecting students to duplicate instruction, conflicting notation, and unmanageable workload.

This document establishes the **Oak Harmonisation Strategy (OHS)**: a unified instructional architecture that unlocks cognitive synergy while strictly safeguarding the statutory integrity of both programmes.

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1. EVALUATION ACROSS THE THIRTEEN PEDAGOGICAL DIMENSIONS

1.1 Content Alignment

- **Status**: Strong fundamental congruence across numbers, algebra, geometry, functions, and data analysis (~75–85% thematic overlap).
- **Strategy**: Establish a **Single Conceptual Core (SCC)** for common topics. Introduce the core mathematical theory once, deeply and rigorously, then branch into programme-specific applications.

1.2 Scope and Sequence Coordination

- **Status**: Occasional sequencing mismatches (e.g., negative numbers in MYP 2 vs. 5ème; vectors in 4ème/3ème vs. MYP 5/DP; sequences in DP 1 vs. Première; logarithms in DP 1 vs. Terminale).
- **Strategy**: Reorder the institutional teaching sequence so that whenever a concept is required by either programme, it is scheduled at the earliest required date. Implement targeted **Bridging Modules (BM)** for early requirements.

1.3 Mathematical Vocabulary (Bilingual Dual-Immersion)

- **Status**: Language barrier (English vs. French) with divergent technical terminology.
- **Strategy**: Adopt a **Systematic Dual-Glossary Method**. Every workbook unit begins with an exhaustive bilingual lexical table displaying exact terms, definitions, false friends, and symbolic equivalents. Instructors use English during IB inquiry sessions and French during APC situation phases.

1.4 Mathematical Notation and Conventions

- **Status**: Divergent conventions: decimal comma (12,5) vs. decimal point (12.5); coordinate notation (A(2 ; 3) vs. A(2, 3)); interval brackets (]a, b[vs. (a, b)); division symbol (: vs. ÷).
- **Strategy**: Explicitly teach notation as a **Context-Dependent Code**. Train students to recognise that both systems represent the identical mathematical reality. Workbooks feature "Notation Alert" sidebars highlighting these variations.

1.5 Conceptual Understanding

- **Status**: Both programmes prioritize conceptual depth (IB via Key/Related Concepts and Global Contexts; Benin APC via contextual problem situations).
- **Strategy**: Fuse the Beninese *Situation de départ* with the IB *Global Context*. Introduce topics through authentic West African and global contexts (e.g., solar energy installations in the Borgou department, shipping logistics at the Autonomous Port of Cotonou, Nokoué lake fisheries, and financial budgeting).

1.6 Procedural Fluency

- **Status**: The Beninese APC curriculum places exceptional value on high-speed, accurate mental arithmetic, hand calculations, algebraic factorization, and polynomial division without calculators. IB assessments incorporate both non-calculator and technology-rich components.
- **Strategy**: Maintain mandatory **"No-Calculator Fluency Drills"** derived from Beninese and IB non-calculator expectations, guaranteeing student speed, number sense, and symbol manipulation stamina.

1.7 Mathematical Reasoning and Deductive Proof

- **Status**: Beninese APC enforces a highly formal deductive structure for geometry (*On sait que... Or si... Donc...*), while IB MYP Criterion B requires inductive pattern discovery leading to general algebraic rules, and Criterion C demands coherent, multi-representational communication.
- **Strategy**: Teach inductive investigation first (explore patterns, generate data, formulate conjectures — IB Criterion B), followed by rigorous deductive proof and formal justification (Benin APC / IB Criterion C).

1.8 Problem-Solving Strategies

- **Status**: Both systems reject passive rote mechanics. Beninese APC uses the *Démarche de résolution de situations-problèmes*; IB uses the *Inquiry Cycle*.
- **Strategy**: Structure classroom problem-solving into a 5-step harmonised protocol:

1. *Understand & Frame*: Define variables, draw diagrams (C_a / Criterion D).
2. *Hypothesize & Conject*: Explore patterns and formulate conjectures (Criterion B).
3. *Select Model*: Choose the theorem, formula, or equation (C_m / Criterion A).
4. *Execute & Calculate*: Perform operations with precision (C_o / Criterion A).
5. *Reflect & Communicate*: Evaluate reasonableness, state units, and justify (CP / Criterion C & D).

1.9 Real-Life Applications and Mathematical Modelling

- **Status**: Rich synergy. Real-world applications are statutory requirements in both systems.
- **Strategy**: Embed dual-purpose contextual tasks in all workbooks. Tasks draw on regional Beninese scenarios (agriculture, microfinance, infrastructure, public health) evaluated alternately through the BEPC/Baccalauréat 3-problem lens and the MYP Criterion D rubric.

1.10 Formative and Summative Assessment

- **Status**: Dual examination expectations: IB MYP Criteria A–D (band 0–8) and DP Papers 1–3 + IA versus Beninese national exams (CEP, BEPC, Baccalauréat) marked on C_a, C_m, C_o, and CP (out of 20).
- **Strategy**: Dual-Track Assessment Architecture. Formative assessments utilize mixed criteria. Summative unit assessments feature two distinct, clearly labelled parts:
- *Part 1: IB Assessment Task* (Criteria A, B, C, or D; or DP Paper style);
- *Part 2: Beninese National Exam Task* (Situation d'évaluation type CEP, BEPC, or Baccalauréat).

1.11 Prerequisite Knowledge and Diagnostic Monitoring

- **Status**: Transfer students or students progressing through the dual stream may encounter prerequisite gaps.
- **Strategy**: Administer diagnostic "Pre-Flight Checks" at the launch of each unit, instantly identifying gaps in fractions, signs, geometry, or algebra, coupled with targeted self-correcting catch-up tasks.

1.12 Cognitive Demand and Progression in Difficulty

- **Status**: Cognitive load must be carefully calibrated to avoid student burnout.
- **Strategy**: Structure all progressive exercises across **Five Calibrated Levels**:
- *Level 1: Foundational Understanding* (definitions, single-step operations);
- *Level 2: Standard Application* (routine multi-step procedural tasks);
- *Level 3: Multi-Step Problem Solving* (combining multiple concepts);
- *Level 4: Reasoning, Investigation & Unfamiliar Contexts* (proofs, non-routine tasks);
- *Level 5: Extension & Programme-Specific Challenges* (advanced national exam problems, DP HL extensions).

1.13 Preparation for Terminal Examinations

- **Status**: High-stakes statutory deadlines:
- Grade 5 (CM2): CEP (Certificat d'Études Primaires).
- Grade 9 (3ème): BEPC (Brevet d'Études du Premier Cycle) and MYP 5 eAssessment.
- Grade 11 (1ère): Baccalauréat Première national benchmarks / DP 2 mock examinations.
- **Strategy**: In CM2, 3ème, and 1ère, the third term integrates dedicated exam-preparation clinics. Students practice under authentic timed conditions with official national scoring grids and official IB mark schemes.

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2. THE OAK HARMONISED INSTRUCTIONAL MODEL

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| THE OAK HARMONISATION ARCHITECTURE                                             |
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1. SHARED CONCEPTUAL FOUNDATION (Co-Taught Core: 60-70% Instructional Time)
- Common mathematical theorems, principles, definitions, and visual models
- Dual-language terminology introduced simultaneously (English & French)
- Joint inquiry into real-world contexts and cross-curricular patterns

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2. IB INQUIRY & CRITERIA TRACK		3. BENINESE APC EXAMINATION TRACK
(15-20% Instructional Time)		(15-20% Instructional Time)
- Criterion B: Pattern Investigation		- Formal deductive proof schemas
- Criterion C: Mathematical Writing		- High-speed non-calculator drills
- Criterion D: Real-life Modelling		- 3-problem Situations d'Évaluation
- GDC Technology & DP Exploration		- Rubrics: Ca, Cm, Co, and CP

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4. BRIDGING & EXTENSION MODULES (As needed for sequencing mismatches)
- Grade 6: Early directed numbers (MYP 2)
- Grade 8: Synthetic vectors & midpoint theorems (Benin 4ème)
- Grade 10: Sequences & Logarithms (DP 1) / 3D Incidence geometry (2nde C)
- Grade 11: Barycentres & Level curves (1ère C) / Integration (DP 2)

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## 3. SPECIFIC CLASS-LEVEL HARMONISATION BLUEPRINTS

### 3.1 Class Level 1: CM2 / MYP 1

- **Harmonisation Thesis**: Combine the rapid calculation agility and commercial arithmetic mastery required by the Beninese CEP with the conceptual geometry and inquiry-based data exploration of MYP 1.
- **Instructional Allocations**:
- Core Shared Units: Four Operations, Fractions, Metric Geometry, Perimeter & Area (65%).
- Beninese APC Priority: Commercial Arithmetic (*prix d'achat, vente, bénéfice*), Agrarian Units (ha, a, ca), CEP exam formatting (20%).
- IB MYP 1 Priority: Coordinate grids, probability scale, Criterion B pattern discovery (15%).

### 3.2 Class Level 2: Sixième / MYP 2

- **Harmonisation Thesis**: Harmonise the deep synthetic geometry of Beninese 6ème (perpendicular bisectors, circles, axial symmetry) with MYP 2 algebra and directed numbers.
- **Instructional Allocations**:
- Core Shared Units: Synthetic plane geometry, axial symmetry, fractions, cubes & prisms (65%).
- Beninese APC Priority: Rigorous deductive proof writing (*On sait que... Or si... Donc...*), non-calculator arithmetic stamina (20%).
- IB MYP 2 Priority: Bridging module on directed numbers (**Z**), Cartesian plane graphing, Criterion B investigations (15%).

### 3.3 Class Level 3: Cinquième / MYP 3

- **Harmonisation Thesis**: High natural alignment. Capitalize on parallel timelines in signed numbers, angles cut by parallel lines, central symmetry, and proportionality.
- **Instructional Allocations**:
- Core Shared Units: Operations in **Z**, parallel angles, triangle angle sum, central symmetry, parallelograms, powers of 10, proportionality (70%).
- Beninese APC Priority: Formal proof of quadrilateral properties, comprehensive descriptive statistics (15%).
- IB MYP 3 Priority: Two-stage probability trees, linear sequence rules, Criterion D applications (15%).
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3.4 Class Level 4: Quatrième / MYP 4

- **Harmonisation Thesis**: Synergise Beninese geometric vector foundations and Pythagoras' theorem with MYP 4 algebraic expansions and multi-ratio trigonometry.
- **Instructional Allocations**:
- Core Shared Units: Pythagoras' theorem, rational numbers (**Q**), exponent laws, algebraic expansion, linear equations, regular pyramids and cones (65%).
- Beninese APC Priority: Synthetic vectors (direction, sense, norm, Chasles), midpoint theorems, triangle centres (20%).
- IB MYP 4 Priority: Full right-triangle trigonometry (sin, cos, tan), cumulative frequency curves and boxplots (15%).

3.5 Class Level 5: Troisième / MYP 5

- **Harmonisation Thesis**: Strategic graduation year. Maintain unified progression across Thales, trigonometry, coordinate geometry, linear systems, and radicals, followed by dual exam clinics.
- **Instructional Allocations**:
- Core Shared Units: Thales' theorem, trigonometry, surd calculations, algebraic identities, linear systems, affine functions, circle theorems (70%).
- Beninese APC Priority: Intensive BEPC workshops: 3-problem contextual evaluations, criterion rubrics (C_a, C_m, C_o, CP) (15%).
- IB MYP 5 Priority: Sine/cosine rules, quadratic formula, MYP Criteria A–D portfolio tasks and eAssessment prep (15%).

3.6 Class Level 6: Seconde / DP 1

- **Harmonisation Thesis**: Integrate the analytical vigor of Seconde C/D (polynomials, quadratics, vector dot products, Al-Kashi, radian trigonometry) with DP 1 functions, sequences, and technology.
- **Instructional Allocations**:
- Core Shared Units: Quadratic equations (Δ), function transformations, 2D vector dot products, Al-Kashi theorem, unit-circle trigonometry (65%).
- Beninese APC Priority: 3D incidence geometry proofs, intervals and absolute value inequalities, polynomial factorisation (15%).
- IB DP 1 Priority: Arithmetic/geometric sequences, logarithms, GDC technology training, bivariate statistics (20%).

3.7 Class Level 7: Première / DP 2

- **Harmonisation Thesis**: Coordinate differential calculus, curve sketching, sequences, and combinatorics, while running distinct parallel tracks for Beninese barycentres and IB integration/IA.
- **Instructional Allocations**:
- Core Shared Units: Differential calculus, curve sketching with asymptotes, numerical sequences, combinatorics and probability distributions (60%).
- Beninese APC Priority: Barycentres of weighted points, partial barycentres proofs, level curves, linear programming (20%).
- IB DP 2 Priority: Anti-differentiation and definite integrals, normal distribution, Internal Assessment Exploration mentorship (20%).

End of Deliverable 4 — Pedagogical Harmonisation Strategy.